24/07/26

convolution is mathematically commutative

downsampling → take an and ignore the rest

upsampling → take an add zero

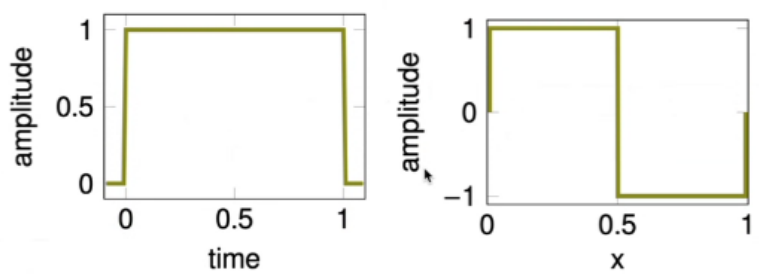
The inverse of mallat's algo. is the inverse of the filter and upsampler

in theory you could do upsampling with the filter inverse and then apply to the signal. So this operation is independent of the signal. This gives us the scaling and wavelet function

Convolution or upsample inverse the signal

As we apply convolution and upsample the signal leads to stacking. Each filter acts a distinct looking signal

- - - Haar Wavelet System - - -

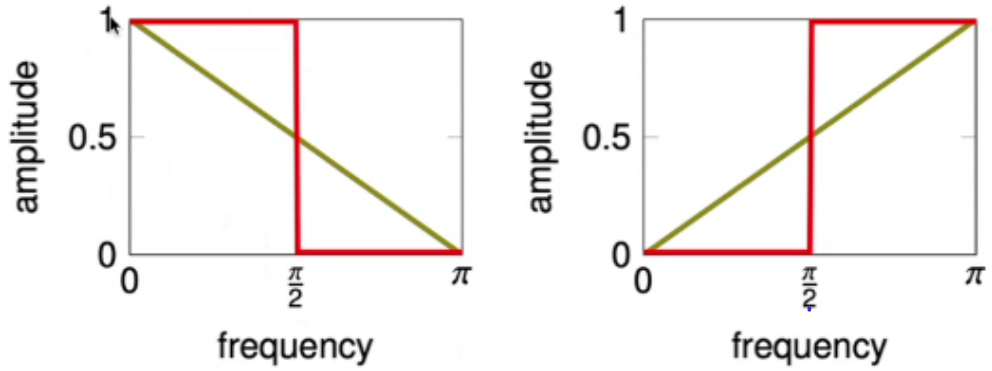


(a) Haar scaling function $\phi(x)$ (b) Haar wavelet function $\psi(x)$

- Haar analysis filters: $h[\cdot] = \{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$ and $g[\cdot] = \{\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\}$.
- Haar synthesis filters: $\bar{h}[\cdot] = \{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$ and $\bar{g}[\cdot] = \{\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\}$.
- The set $\{h[\cdot], g[\cdot], \bar{h}[\cdot], \bar{g}[\cdot], \phi(x)$ and $\psi(x)\}$, containing six elements, defines de Haar wavelet system.
- Curiously, for the Haar wavelet system, $h[\cdot] = \bar{h}[\cdot]$ and $g[\cdot] = \bar{g}[\cdot]$.

filter with size 2 is not good in frequency

- - - Haar Wavelet System - - -



(c) in olive: frequency response for $h(\cdot)$; in red: ideal response of a half-band low-pass filter

(d) in olive: frequency response for $g(\cdot)$; in red: ideal response of a half-band high-pass filter

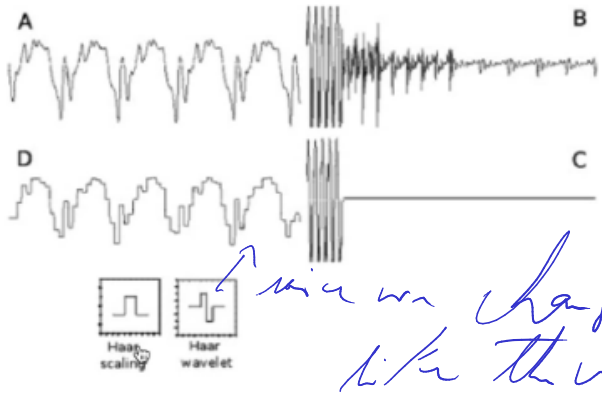
We can do lossless or lossy compression

inverse = original

can be some stuff

to compress a signal we could move holes with not much energy. To uncompress we fill the other holes with zeros and do the inverse, giving an approx. of $S[\cdot]$

- - - Example Application: lossy compression - - -



compressed

since we changed a lot it looks like the wavelet function

- In the figure above we can see: (A): original signal; (B): transformed signal in level three; (C): transformed signal modified in such a way that only the trend in the third level remains intact; (D): reconstructed signal. Observe how the reconstructed signal shape is similar to the shapes of the scaling and wavelet functions, dilated and translated.

To make it better, we could choose different filters to make the wavelet function look more like the original

Filters of Daubechies filter bank, a perfect Reconstruction filter bank, $g[\cdot]h[\cdot]$ do and $\bar{g}[\cdot]\bar{h}[\cdot]$ made exactly they are in different signs starting with 4. They are always even sized

wavelet better

- - - Wavelet Families: Daubechies - - -

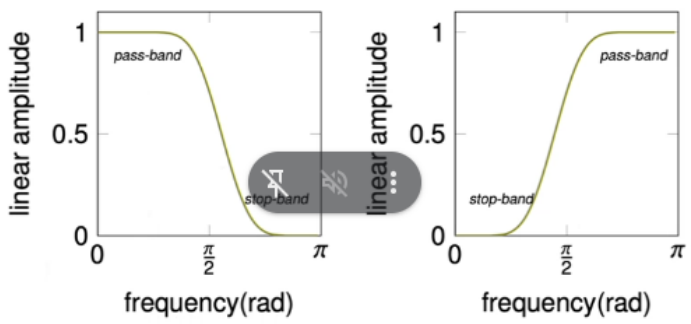


Figure: left and right, respectively: the frequency responses for Daubechies filters $h[\cdot]$ and $g[\cdot]$ with support-size $N = 16$. The higher N is, the more the frequency responses get closer to the ideal responses. The flat behaviour in the pass- and in the stop-bands is an intrinsic characteristic of Daubechies filters. In terms of frequency response, Daubechies filters are much better than Haar filters.

Daubechies filters are known to be maximally flat because in max amplitude they keep flat

how they were constructed

- Daubechies filters are specified as follows:
  - Force $g[\cdot]$ to have unitary energy so that the DTWT preserves that of the input signal, i.e.,

$$\sum_{k=0}^{N-1} g_k{}^2 = 1$$

← filters $h[\cdot]$ and $g[\cdot]$ have unitary energy.

  - Impose $\frac{N}{2}$ vanishing moments on the wavelet function, i.e.,

$$\sum_{k=0}^{N-1} g_k k^b = 0 \qquad ,(b = 0, 1, ..., \frac{N}{2} - 1)$$

→ polynomial interpolation

  - Define $\frac{N}{2} - 1$ orthogonality conditions related to the translations of the filter so that the transformation matrix used to carry out Mallat's algorithm is orthogonal, allowing for signal re-synthesis based on its transpose:

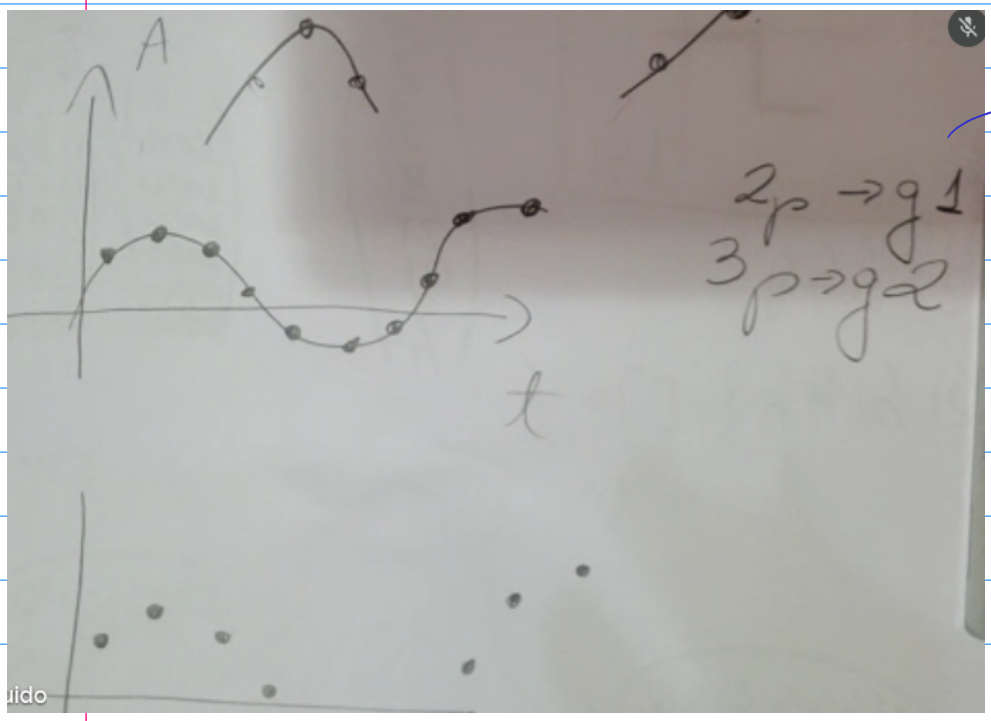
$$\sum_{k=0}^{N-1} g_k g_{k+2l} = \delta_{0,l} \qquad ,$$

← orthogonal

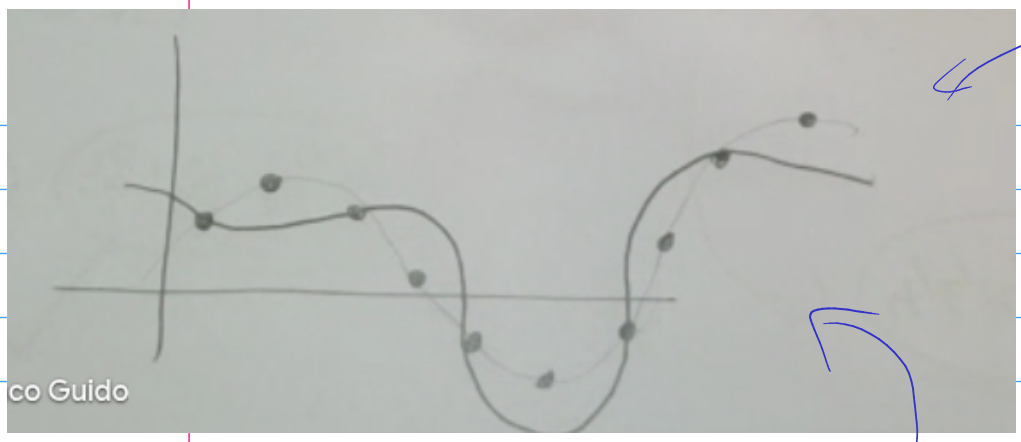
being δ the Dirac delta and $l \in \mathcal{Z}$;

90 / 200

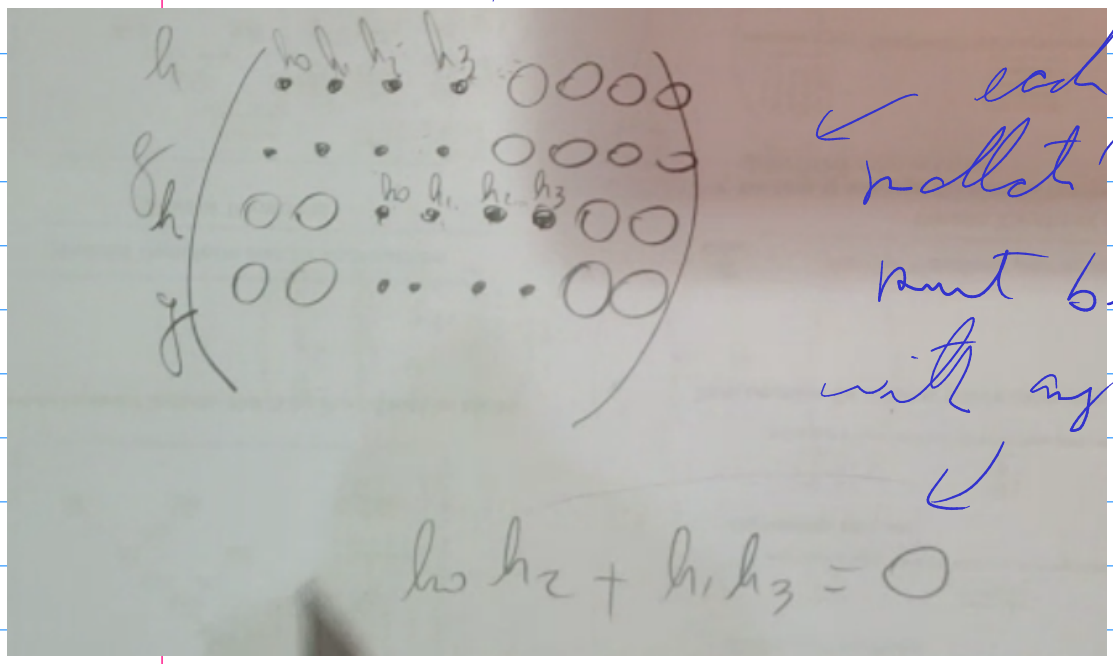
unitary filters preserve the signal energy



→ the non part the polinomial interpolation needs n-1 degree polynomy



to filter this signal we could take as point and ignore the point giving a polynomial interpolation 1/2



each line is mallat's algo. must be altogether with any other

- The equations are grouped together, resulting in a non-linear system of N equations in N unknowns. Then, the system is solved using any iterative numerical procedure, such as Gauss-Siedel's, Jacobi's or Newton's methods, to obtain the high-pass filter $g[\cdot]$. Lastly, the filter $h[\cdot]$ is obtained so that $h_k = (-1)^{k+1} g_{N-k-1}$, in order to complete the analysis filter pair. The filters $\bar{h}[\cdot]$ and $\bar{g}[\cdot]$ can also be obtained, in such a way that $\bar{h}_k = h_{N-k-1}$ and $\bar{g}_k = (-1)^{k+1} h_k$ characterize the synthesis filter pair. Lastly, the corresponding scaling function,

$$\phi(x) = \sum_k h_k \phi(2N - k)$$

and the wavelet function

$$\psi(x) = \sum_k g_k \phi(2N - k)$$

can also be obtained.

a wavelet or Daubechies find a numerical approach to obtain the function

- **Numerical Example**: for $N = 4$, we have the following nonlinear system of $N = 4$ equations in $N = 4$ unknowns:

$$\begin{cases} (g_0)^2 + (g_1)^2 + (g_2)^2 + (g_3)^2 & = & 1 & \text{(filter energy)} \\ g_0 + g_1 + g_2 + g_3 & = & 0 & \text{(first vanishing moment)} \\ 0g_0 + 1g_1 + 2g_2 + 3g_3 & = & 0 & \text{(second vanishing moment)} \\ g_0 g_2 + g_1 g_3 & = & 0 & \text{(required to force orthogonality)} \end{cases}$$

Its solution corresponds to the Daubechies filter $g[\cdot]$ with support-size $N = 4$, i.e., $g[\cdot] = \{-0.1830127018, -0.3169872981, 1.1830127018, -0.6830127018\}$. Consequently, $h[\cdot] = \{0.6830127018, 1.1830127018, 0.3169872981, -0.1830127018\}$. To ease calculations by hand, we can also, equivalently, use the following format for the solutions: $h[\cdot] = \{\frac{1+\sqrt{3}}{4\sqrt{2}}, \frac{3+\sqrt{3}}{4\sqrt{2}}, \frac{3-\sqrt{3}}{4\sqrt{2}}, \frac{1-\sqrt{3}}{4\sqrt{2}}\}$ and $g[\cdot] = \{\frac{1-\sqrt{3}}{4\sqrt{2}}, \frac{-3+\sqrt{3}}{4\sqrt{2}}, \frac{3+\sqrt{3}}{4\sqrt{2}}, \frac{-1-\sqrt{3}}{4\sqrt{2}}\}$.
- Although we do **not** need them to perform DTWTs, the scaling and wavelet functions can also be calculated explicitly or by iteration.

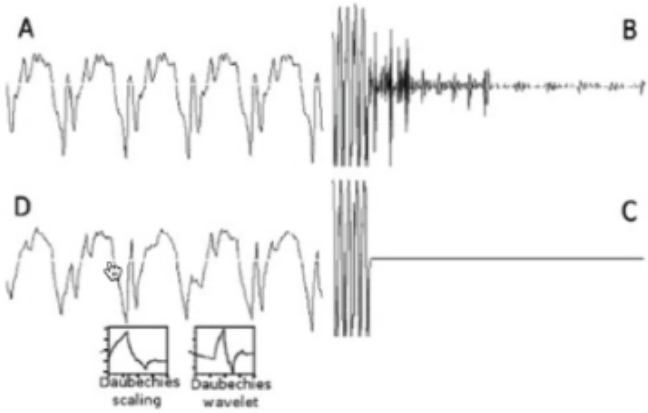
- The scaling function, $\phi(x)$, defined recursively using the dilation equation $\phi(N) = \sum_k h_k \phi(2N - k)$ for a system of support $N = 4$, does not exist outside the interval $[0 - 3]$. We therefore get:

$$\begin{cases} \phi(0) = h_0\phi(0) \\ \phi(1) = h_0\phi(2) + h_1\phi(1) + h_2\phi(0) \\ \phi(2) = h_1\phi(3) + h_2\phi(2) + h_3\phi(1) \\ \phi(3) = h_3\phi(3) \end{cases}$$

or $MT = T$, being $M = \begin{pmatrix} h_0 & 0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ 0 & h_3 & h_2 & h_1 \\ 0 & 0 & 0 & h_3 \end{pmatrix}$ and $T = \begin{pmatrix} \phi(0) \\ \phi(1) \\ \phi(2) \\ \phi(3) \end{pmatrix}$. So, matrix T with scaling function values is the eigenvector of M corresponding to eigenvalue 1. Using the normalizing condition $\sum_k \phi(k) = 1$, we get:

$$\begin{cases} (h_0 - 1)\phi(0) = 0 \\ h_2\phi(0) + (h_1 - 1)\phi(1) + h_0\phi(2) = 0 \\ h_3\phi(1) + (h_2 - 1)\phi(2) + h_1\phi(3) = 0 \\ (h_3 - 1)\phi(3) = 0 \\ \phi(0) + \phi(1) + \phi(2) + \phi(3) = 1 \end{cases} \overset{(\bullet)}{\Rightarrow} \begin{cases} \phi(0) = \phi(3) = 0 \\ (h_1 - 1)\phi(1) + h_0\phi(2) = 0 \\ h_3\phi(1) + (h_2 - 1)\phi(2) = 0 \\ \phi(1) + \phi(2) = 1 \end{cases},$$

- - - Example Application Revisited: lossy compression - - -



using these new filters it looks much like the signal s[·]

- This is the same lossy compression application we studied in slide 86 but now using Daubechies filters of support-size 4 instead of Haar. Observe the differences... Since the corresponding scaling and wavelet functions of Daubechies filters with support-size 4 are smoother, the reconstructed signal in (D) is much better than that obtained with Haar... Please, compare with slide 86.

Mallat's algorithm by default throws points away, but it's small. Subtleties don't lift that, and there is another formulation, stationary WT that reverses the downsampling

Vanishing moments → means that our signal points is an interpolation